

YOUNGRAE KIM

youngrae@usc.edu youngraekimm.github.io

EDUCATION

University of Southern California (USC)

Ph.D. in Computer Engineering (Advisor: Prof. Jay Kuo, Prof. Peter Beerel)
– Cumulative GPA: 3.8 / 4.0

Los Angeles, CA, USA
Aug. 2024 – Present

Korea Advanced Institute of Science and Technology (KAIST)

M.S. in Computer Science (Advisor: Prof. Dongman Lee)
– Thesis: "Few-Shot Weather-Degraded Image Restoration"
– Cumulative GPA: 3.62 / 4.0

Daejeon, Korea
Feb.2022 – Feb.2024

Hongik University

B.S. in Computer Engineering (Advisor: Prof. Young Yoon)
– Major GPA: 98.41%; Cumulative GPA: 95.10%

Seoul, Korea
Mar.2016 – Feb.2022

RESEARCH INTERESTS

My research interests primarily lie in **Video Generation/Understanding**, **Multimodal LLMs**, and **Domain Generalization**.

RESEARCH EXPERIENCE

Google

Student Researcher, Mentor: Srinivas Kaza, Xuan Luo

May. 2026 – Present

USC MCL Lab

Graduate Research Assistant, Advisor: Prof. Jay Kuo

Aug. 2024 – Present

USC EESSC Lab

Graduate Research Assistant, Advisor: Prof. Peter Beerel

Aug. 2024 – Present

Kyung Hee University VLL Lab

Research Collaborator, Advisor: Prof. Jinwoo Choi

May 2023 – May 2026

KAIST CDSN Lab

Graduate Research Assistant, Advisor: Prof. Dongman Lee, Prof. Seunghoon Hong

Feb. 2022 – Jul. 2024

Hongik University APL Lab

Research Intern, Advisor: Prof. Young Yoon

Jul. 2020 – Dec. 2021

PUBLICATIONS

P4. MemRoPE: Training-Free Infinite Video Generation via Evolving Memory Tokens

Youngrae Kim*, Qixin Hu*, C.-C. Jay Kuo, Peter Beerel

Pre-Print

P3. Perturbation-Aware Training and Hardware Chopping for Robust Analog Neural Network Accelerators

Maysara Hamada*, **Youngrae Kim***, Sreetama Sarkar, Chee-An Yu, Jake Hyun, Huixuan Yin, Hsiang-Chun Cheng, Jian Meng, Yifan He, Jae-sun Seo, C.-C. Jay Kuo, Peter Beerel, and Chen Shuo-Wei

Pre-Print

P2. Beyond the Mean: Three-Axis Fidelity for Aligning LLM-Based Survey Simulators from Small Pilot Data

Eun Cheol Choi*, **Youngrae Kim***, Prabhu Pugalenthi*, Hong-En Chen*, Bo-Ruei Huang*

Pre-Print

P1. Rethinking Layer Redundancy in Large Language Models: Calibration Objectives and Search for Depth Pruning

Minkyu Kim*, Vincent-Daniel Yun*, **Youngrae Kim***, YoungJin Heo*, Suin Cho*, Seong-hun Kim*, Woosang Lim*, Gaeul Kwon*

Pre-Print

- C6. **SlotVTG: Object-Centric Adapter for Generalizable Video Temporal Grounding**
 Jiwook Han*, Geo Ahn*, **Youngrae Kim***, Jinwoo Choi
GRAIL-V Workshop at IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) 2026
- C5. **Feature Augmentation based Test-Time Adaptation**
 Yeonggeol Cho*, **Youngrae Kim***, Junho Yoon, Seunghoon Hong, Dongman Lee
IEEE/CVF Winter Conference on Applications of Computer Vision (WACV) 2025
- C4. **DEVIAS: Learning Disentangled Video Representations of Action and Scene**
Youngrae Kim*, Kyungho Bae*, Geo Ahn*, Jinwoo Choi
European Conference on Computer Vision (ECCV) 2024 – (Oral Presentation, 2.3% acceptance rate)
- C3. **MetaWeather: Few-Shot Weather-Degraded Image Restoration**
Youngrae Kim*, Yeonggeol Cho*, Thanh-Tung Nguyen, Seunghoon Hong, Dongman Lee
European Conference on Computer Vision (ECCV) 2024
- C2. **Beyond Entropy: Style Transfer Guided Single Image Continual Test-Time Adaptation**
 Yeonggeol Cho*, **Youngrae Kim***, Dongman Lee
Workshop on Vision-Centric Autonomous Driving at European Conference on Computer Vision (ECCV) 2024
- C1. **Efficient Reference-based Video Super-Resolution (ERVSR): Single Reference Image is All You Need**
Youngrae Kim*, Jinsu Lim*, Hoonhee Cho*, Minji Lee*, Dongman Lee, Kuk-Jin Yoon, Ho-Jin Choi
IEEE/CVF Winter Conference on Applications of Computer Vision (WACV) 2023 – (Oral Presentation)
- J1. **Zone-Agnostic Greedy Taxi Dispatch Algorithm Based on Contextual Matching Matrix for Efficient Maximization of Revenue and Profit**
Youngrae Kim, Young Yoon
Electronics 2021
 (* denotes equal contribution)

PATENTS

- Dongman Lee, Yeonggeol Cho, **Youngrae Kim**, Junho Yoon. Korean Patent, Application No. 10-2025-0066479. Single Image Based Style Transfer Guided Continual Test-Time Adaptation System. 2025.

FELLOWSHIPS AND HONORS

Outstanding Poster Presentation Award, Workshop on Image Processing and Image Understanding (IPIU)	Feb. 2026
Oral Presentation, European Conference on Computer Vision (ECCV)	Sep. 2024
Best Paper Award, Korea Computer Congress	Jun. 2024
Honorable Mention, Workshop on Image Processing and Image Understanding (IPIU)	Feb. 2024
Full-tuition Government Scholarship for Science and Engineering, KAIST	2022 – 2024
Oral Presentation, IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)	Jan. 2023
Best TA Award, KAIST	Fall 2022
Academic Excellence Scholarship for 7 semesters, Hongik University	2016 – 2022

ACADEMIC SERVICES

Invited Talk: Toward Infinite Video Generation, eBay (host: Ashirbad Mishra), May 2026
Reviewer: CVPR 2024, 2025, 2026; ECCV 2026; NeurIPS 2026; AAAI 2025, 2026; WACV 2025; BMVC 2026; TCSVT

TEACHING

<i>Teaching Assistant</i> , CS206 Data Structure, KAIST	Fall 2022
<i>Teaching Assistant</i> , CS330 Operating System, KAIST	Spring 2022

LEADERSHIP EXPERIENCE

<i>Vice President</i> , Korean Graduate Student Association (KGSA), USC	May 2026 – Present
<i>Troop Leader</i> (Sergeant, Honorable Discharge), Republic of Korea Air Force	Apr. 2018 – Mar. 2020